INTERIM FEDERAL SPECIFICATION

FLASHLIGHT, TRAFFIC

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a tubular shape hand flashlight with a removable luminous wand used for the control and directing of vehicle and pedestrian traffic.

1.2 Classification.

1.2.1 Types, classes, and styles. The flashlight furnished under this specification shall be of the following types, classes, and styles, as specified (see 6.1).

Types:
I - Metal case
II - Plastic case

Class:
A - 8 inch wand
B - 5 inch wand
C - 3 inch wand

Style:
1 - Red wand
2 - Yellow wand

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

L-P-390 - Plastic Molding Material, Polyethylene, Low and Medium Density
L-P-396 - Plastic Molding, Polystyrene
L-P-397 - Plastic Molding Material, Cellulose Acetate
W-B-101 - Battery, Dry
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-601 - Boxes, Woodcleated-Plywood
PPP-B-621 - Boxes Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Fiber
PPP-B-640 - Boxes, Corrugated, Triple Wall
PPP-B-665 - Boxes, Paperboard, Metal Stayed
PPP-B-676 - Boxes, Set-up, Paperboard

Federal Standards

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)
Fed. Std. No. 595 - Colors

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is on sale on the subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20406.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo.

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Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.
(Federal Government activities may obtain copies of Federal Specification, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:
MIL - P- 116 - Preservation, Methods of
MIL - P- 22985 - Plastic, Ethyl Cellulose
MIL - C- 25050 - Colors, Aeronautical Lights and Lighting Equipment, General Requirements for

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society For Testing and Materials (ASTM):
D 1248-65T - Polyethylene Molding and Extrusion Materials
D 1788-62T - Rigid Acrylonitrile - Butadiene - Styrene (ABS) Plastic

(Copies may be obtained from the American Society for Testing and Materials 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General. The flashlights furnished under this specification shall be of tubular construction consisting of a plastic wand, reflector, lens, retaining ring, end cap, case, lamp, lampholder, switch, and suspension ring, as specified herein. The flashlights shall be designed to operate on "D" type dry cell batteries that are in accordance with W-B-101. Batteries shall not be furnished.

3.2 Material. The materials used throughout shall be suitable in every respect for their intended purpose and such that the finished components shall evidence no indication of damage, protective coating penetration, or corrosion when the flashlight is subjected to the tests specified in section 4. Dissimilar metals shall not be used in intimate contact with each other unless protected against electrolytic action. Switch contacts shall be of commercial grade or better spring brass phosphor bronze or beryllium copper. All other electrical contacts may be constructed of the above stated material or of copper coated steel. The top and bottom contact strips shall meet the requirements of this specification and shall be in accordance with good commercial standards for the intended use. End or base-cap springs may be of a suitable material, such as copper steel, provided the material used has suitable electrical conductivity charateristics and meets all other requirements of this specification.

3.2.1 Wand. The wand shall be fabricated from a polyethylene material, conforming to L-P-390 for the medium desity and ASTM D1248-65T for High desity. The wands shall be colored in accordance with Federal Standard No. 595, Style 1 shall be aviation Red No. 21105 and style 2 shall be aviation yellow No. 23538. Light transmission charateristics shall be in accordance with MIL-C-25050.

3.2.2 Reflection. The contour of the reflector shall be essentially parabolic. The reflecting surface shall be smooth and free from bubbles and pits.

3.2.3 Lens. The lens shall be formed from clear, cellulose acetate plastic in accordance with L-P-397 or polystyrene plastic in accordance with L-P-396, Type 1.

3.2.4 Light projection. When tested in accordance with 4.4.2, the flashlight shall give illumination equal to or higher than 150 for the eight (8) inch circle (4 points) and 350 for the center and four (4) inch circle (5 points).

3.2.5 Case.

3.2.5.1 Type I. The type I metal shall be either chrome plated copper coated steel or aluminum of sufficient thickness to meet the requirements of this specification and in accordance with the best commercial practices for the intended use. The case may be manufactured with or without the threaded end cap. In either case the suspension ring shall be included and shall meet all requirements of this specification.

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3.2.5.2 Type II. The type II plastic case with end cap and retaining ring shall be molded of plastic conforming to specification MIL-P-22985 or ABS plastic in accordance with ASTM D1788. The color shall be of the manufacturers option unless otherwise specified.

3.3 Construction.

3.3.1 Caps and retaining rings. Means shall be provided through the use of threaded caps or retaining rings for securing the lens and batteries. The rings and caps shall have a minimum of three engaged threads for the type II and a minimum of two engaged threads for the type I.

3.3.2 Threads. All threads shall be of design and fit that may be screwed tight using a torque not exceeding 25 inch pounds. Threads shall not strip or ride over those of a mating part when a torque of 30 inch pounds is used.

3.3.3 Dimensions. The dimensions of flashlights shall be in accordance with the values shown in table I. The effective inside diameter of the battery compartment shall be not less than 1-3/8 inch nor the maximum diameter more than 1-1/2 inches, and its length and the design of the rear contact spring shall be such that it will accomodate the required number of cells of any length between 2-5/16 inches and 2-13/32 inches. The spring shall compress a minimum of 3/8 inch and exert a force of not less than two pounds when tested in accordance with 4.4.5.

TABLE I

Class	No. of Cells	Industry Lamp No.	Max. Overall Dimensions in inches including wand		Wand Lg from lens	Wand wall thk.	Wand aperture small open end
			Length -3/4" +1.00	Diameter -1/4 +1/4	-3/4" +1.00	-0 +1/16	- 1/4" + 1/4"
A	2	PR 2	15-1/2	2-1/8	8	1/16	1-3/8
B	2	PR 2	11-1/2	2-1/8	4-3/4	1/16	1-3/8
C	2	PR 2	9-1/2	2-1/8	2-1/2	1/16.	1-1/2

3.3.4 Lamp and lampholders. Lamps shall be miniature flanged base single contact and shall be solidly retained in place in the lamp receptacle. Means shall be provided to prevent the dry cells from damaging the base of the lamp when tested in accordance with section 4. Dry cells shall be held in place against the lampholder by a helical spring or other acceptable means. Lamps supplied with the flashlights shall be prefocused miniature tungsten filament of the commercial type as specified in table I.

3.3.5 Switch. The switch shall be mounted on the side of the case and shall be a slide type that will be solidly retained in position when placed in either the on or off positions. When specified, (see 6.1) a "flashing"position shall be provided. Switches shall resist accidental movement to the "on" position. The switch slide or button shall be designed so as to permit easy manipulation with one hand. The design and construction of the switch mechanism shall be such that it will operate in a satisfactory manner for 25,000 cycles at a rate of 15 cycles per minute (see 4.4.7.) The construction shall be such that with the flashlight completely assembled, fresh dry cells in place and the switch in the off position, it shall be impossible to energize the lamp by bridging any two points on the exposed surface with an electrical conductor.

3.3.6 Suspension Ring. A suspension ring shall be secured to the bottom cap and fastened in such a manner that it cannot be readily detached by hand. The suspension ring arrangement shall be so designed that the ring will seat firmly to and recess in the end cap. The suspension ring shall be capable of supporting without distortion a weight of 15 pounds.

3.3.7 Roll. The flashlight fully assembled shall be so constructed that when placed in a horizonal position on a stationary flat surface inclined at an angle of 5 degrees will not roll more than 4-½ inches.

3.3.8 Interchangeability. All corresponding parts of flashlights of the same types, style, size and manufacture shall be mechanically and electrically interchangeable.

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3.3.9 Cleaning. All parts shall be throughly cleaned. Traces of corrosive compounds used for cleaning, plating, finishing or machining shall be removed or chemically neutralized to prevent subsequent corrosion.

3.3.10 Workmanship. The flashlight shall be fabricated and finished in a thoroughly workmanship-like manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; marking and dimensions; thoroughness of soldering, painting, and riveting, alignment of parts and tightness of assembly screws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at room temperature, pressure, and humidity.

4.2 Sampling.

4.2.1 Visual and mechanical. Sampling for visual and mechnical inspection shall be conducted in accordance with MIL-STD-105, inspection level S4, acceptable quality level (AQL) 2.5.

4.2.2 Tests. Sampling for test shall be conducted in accordance with MIL-STD-105, inspection level S3, (AQL) 2.5.

4.3 Inspection. A visual and mechnical inspection shall be conducted in accordance with, but not limited to, the following classification of defects.

Visual and Mechnical Inspection

Materials, parts, or accessories, missing or not as specified.
Dimensions not as specified.
Threaded parts not as specified.

4.4 Tests

4.4.1 Wand mounting. The wand mounting, excluding the class C, shall be tested as follows. The wand shall be assembled to the flashlight, and the flashlight shall be securely mounted to a table or stand parallel to the floor The flashlight shall be secured, so that the wand will extend beyond the edge of the table or stand. A weight shall be suspended 1/4-inch from the small end of the wand. For the Class A an eight pound weight shall be used. For the Class B, a ten pound weight shall be used. The wand shall remain intact without crazing or pulling away from the flash light for a minimum of 2 minutes, (See 3.2.1)

4.4.2 Illumination readings. Illumination readings shall be taken with assembled flashlight including lens, bulb, and new batteries but without wand. The flashlight shall be arranged so that its beam is projected on a screen, the plane of which is perpendicular to the optical axis of the flashlight with ten (10) feet between the lens and the screen as described below. For test purposes, the geometric center of the projected beam of light may be considered the point at which the optical axis of the flashlight pierces the test plane. Concentric circles one of 8-inch diameter and one of 4-inch diameter shall be drawn on the screen.

The intersection of these circles with their vertical and horizontal diameters shall be indicated. The intensity of the light falling on the screen at the 8 points of intersection and at the center of the circles shall be read using a light sensitive element having a diameter of 1.5 ± 0.05 inches, centered on the points. The flashlight shall be turned on for 15-second intervals, and the reading for a point shall be taken at the end of the time interval. The flashlight will be turned off for 30 seconds between readings. The center point will be read first, followed by the points on the inner circle and the points on the outer circle.

Conversion formula: Candle power = 100 X meter reading in foot candles. The light measuring equipment shall have a known accuracy of ± 5 percent or better.

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4.4.3 Drop Test. The assembled flashlight, complete with dry cells, wand, lamps and lens, shall be dropped in free fall through a distance of 8 feet on the center of a bed of loose dry sand. The sand shall be at least two feet by two feet by six inches deep and the exposed surface shall be smoothed to a horizontal plane prior to each drop. Each flashlight shall be dropped so that it strikes ten times in an approximate horizontal position, ten times in a vertical position upon the wand, and ten times in a vertical position upon the base of the flashlight. Tightening of threaded parts shall not be made once the test has started. At the completion of the tests flashlights shall remain intact, without cracks or damage to any part (see 3.2.5).

4.4.4 Thread Test. Starting with the case of the flashlight, disassembled, it shall be assembled, applying a measured torque of ten-inch pounds ± 1/2-inch pound, to "Screw-Up" the caps and other threaded parts. The flashlight shall be examined to determine compliance with requirement 3.3.2. A measured torque of 25-inch pounds ± 1/2-inch pound shall be applied to each of the threaded parts of the case and the flashlight shall then be examined for compliance with the stated requirements of 3.2.2.

4.4.5 Battery spring tests. With the lens ring, reflector, wand, and lampholder removed, the flashlight resting on its base, and the required number of dry cells in place, a force shall be applied to the top of the upper dry cell as necessary to depress the spring and move dry cells relative to the case to the position which they will occupy when the flashlight is completely assembled. Spring extension and strength requirements shall be as specified in 3.3.3.

4.4.6 Dimensional and material composition stability test. One half of the flashlights in a sample completely assembled, including batteries, shall be subjected to a temperature of 160° ± 2°F dry heat for 16 hours, followed by a temperature of 160° ± 2°F. at 90 to 100 percent relative humidity for six hours. The flashlights shall then be compared with untested flashlights for dimensional stability, crazing of surface, changes in appearance of materials and other defects. The parts of tested samples shall be interchangeable with corresponding parts of untested samples and after new batteries have been installed, the test samples shall be operable (see 3.3.3).

4.4.7 Switch. Flashlights shall be tested as follows: The sustained contact switch mechanism of the flashlight shall be tested by operating the button for 25,000 cycles at a rate of 15 cycles per minute. A cycle shall consist of movements from "OFF" position to full "ON" position and back to "OFF" position. The switch shall be operated under normal electrical load conditions, and the lamp and batteries shall be renewed as often as required to insure that the switch mechanism is operating under normal load through out the 25,000 cycles (see 3.3.5). Upon completion of the above test the flashlight shall function properly and the light shall not go off when shaken vigorously with the switch in the "ON" position.

4.4.8 Material test. After completion of the switch test, the illumination reading test using the same meter shall be run again, only on those points located on the four (4) inch circle and in the center (5 points). The result shall be compared with the previous test and there shall not be a deviation of more than 7 percent from the previous reading. New batteries and bulbs may be used for this test.

4.4.9 Suspension Ring. The suspension ring shall be tested by suspending a weight of 15 pounds from the ring for a period of 1 minute with the flashlight in a vertical position, base down, to determine compliance with 3.3.6. After this test the ring shall show no sign of distortion.

4.4.10 Impact test. The flashlight shall be subject to a temperature of 0° ± 5°F, for 2 hours and shall then be subjected to a 12-inch pound impact using a one pound steel ball. The impact shall be applied (a) to the outside of the case tube at a point on the side opposite the switch, (b) to the wand and to the bottom cap. Failure to operate normally or any chipping or cracking shall constitute failure of this test.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.5 defects per hundred units.

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Table II. Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

5. Preparation FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The flashlight shall be preserved and packaged individually by method 1A8 of MIL-P-116 in containers conforming to PPP-B-566, PPP-B-676 or PPP-B-665.

5.1.2 Level B. Each flashlight shall be packaged in a close-fitting paperboard box conforming to PPP-B-566, PPP-B-665 or PPP-B-676.

5.1.3 Level C. The flashlights shall be packaged to afford adequate protection against deterioration and damage during shipment from the supplier to the initial destination.

5.1.4 Intermediate packaging. Twelve unit packages of flashlights shall be packaged in a close-fitting paperboard or fiberboard box conforming to PPP-B-566, PPP-B-665, PPP-B-676 or PPP-B-636, grade 175, class domestic.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Flashlights preserved and packaged as specified in 5.1 in quantities as specified (see 6.2) shall be packed in export type shipping containers conforming to PPP-B-636, PPP-B-640, PPP-B-621, or PPP-B-601 as applicable to weight and size limitations. Closure shall be in accordance with the requirements of the appendix to the container specification.

5.2.2 Level B. Eight intermediate packages of flashlights packaged as specified in 5.1.2 and 5.1.4 shall be packed in a fiberboard box conforming to PPP-B-636, class domestic. The box selected shall meet the special requirements of table I of PPP-B-636. Closure shall be in accordance with the appendix to PPP-B-636.

5.2.3 Level C. Flashlights shall be packed to insure carrier acceptance and safe delivery at desdination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order the interior packages and the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

a. Title, number and date of this specification.
b. Type, class and style (see 1.2.1).

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6.1 Ordering data. (Continued)

c. Switch with flashing position if required (see 3.3.5).
d. For plastic case flashlights optional color (see 3.2.5.2).
e. Selection of the applicable level of packaging and packing required (see 5.1 and 5.2).
f. Quantity of unit packages required in the shipping container (see 5.2.1).

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